

Lottery Numbers

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DSC640: Data Presentation & Visualization

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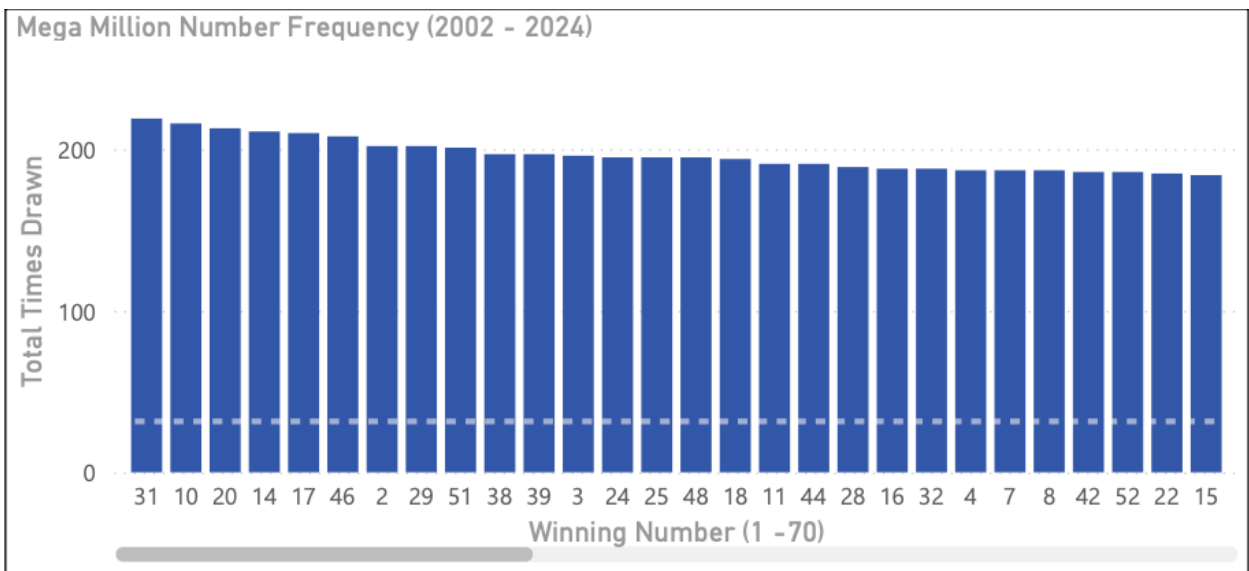
The Equilibrium of Chance: A Statistical Review of NY Lottery Trends

Audience, Purpose, and Medium

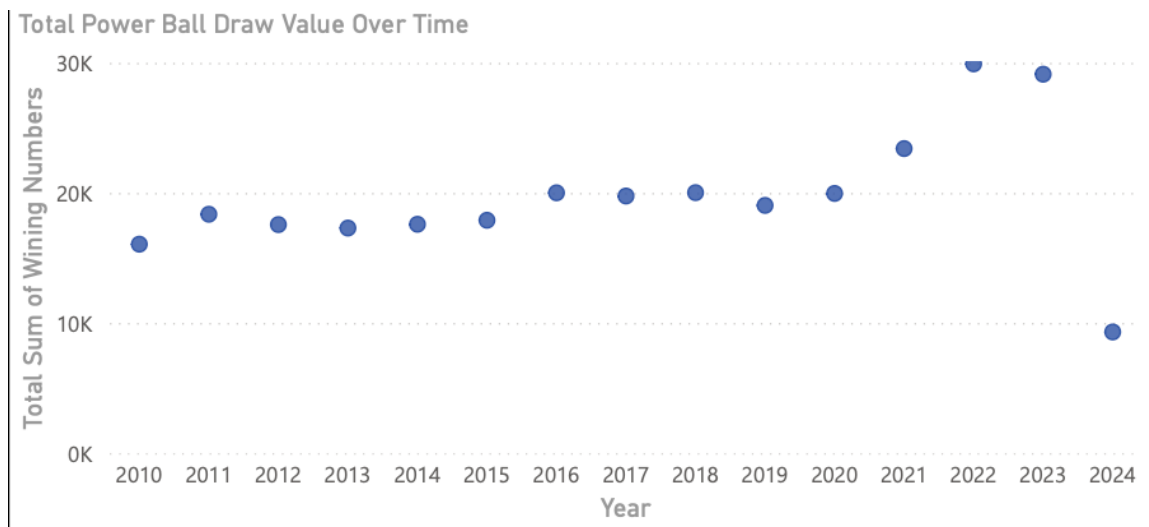
This analysis is designed for the general public—specifically regular lottery participants—to demystify the statistical reality of "lucky" numbers. The purpose is a Call to Action: I intend for the audience to move away from viewing the lottery as a financial strategy and instead treat it as a small entertainment expense. The goal is to encourage redirecting a portion of "lotto funds" into mathematically sound investment vehicles where the probability of growth is a certainty rather than a miracle. The medium is a Power BI Dashboard, chosen for its ability to transform thousands of rows of historical data into a single, cohesive narrative of randomness.

The Story of the Data

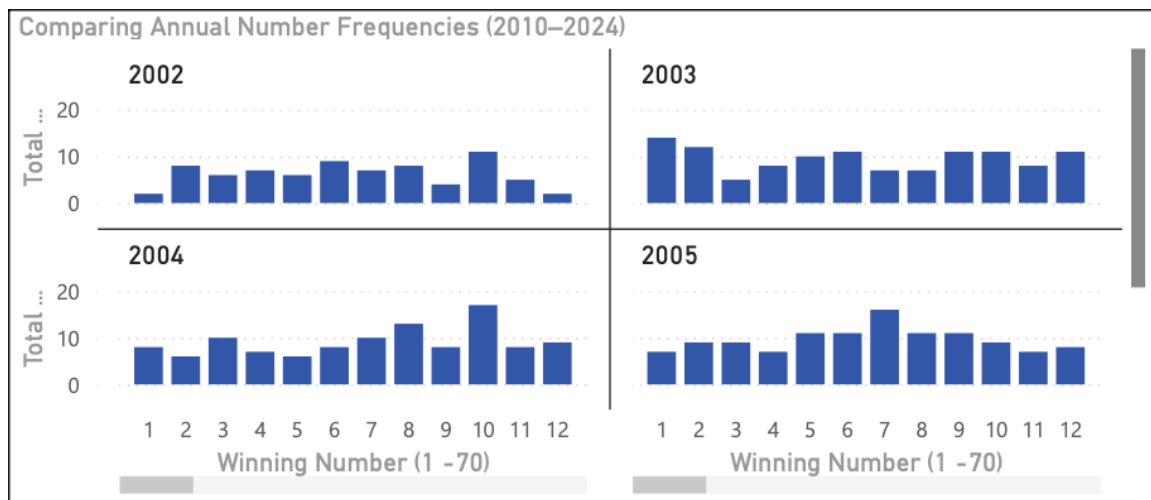
- 1. **Histogram (Mega Millions Frequency):** Represents Mathematical Equilibrium. By showing that numbers 1–70 appear with nearly identical frequency over 20 years, it visually debunks the myth of "hot" or "cold" numbers.



- 2. **Scatterplot (Randomness Index):** Represents **Consistent Chaos**. By plotting the total sum of each draw over time, it shows a stable "cloud" with no trendline, proving that today’s draw is just as unpredictable as those from a decade ago

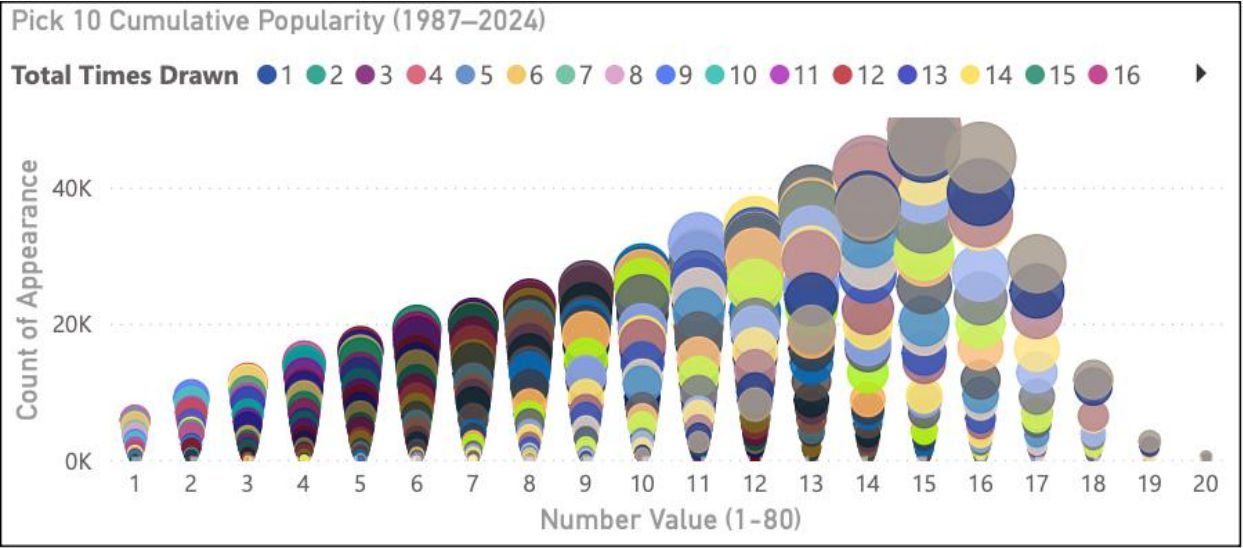


3. **Small Multiples (Yearly Trends):** Represents the **Transience of Luck**. side-by-side yearly grids show that a number that peaked in 2015 often disappeared in 2016, proving that patterns are merely temporary statistical noise.

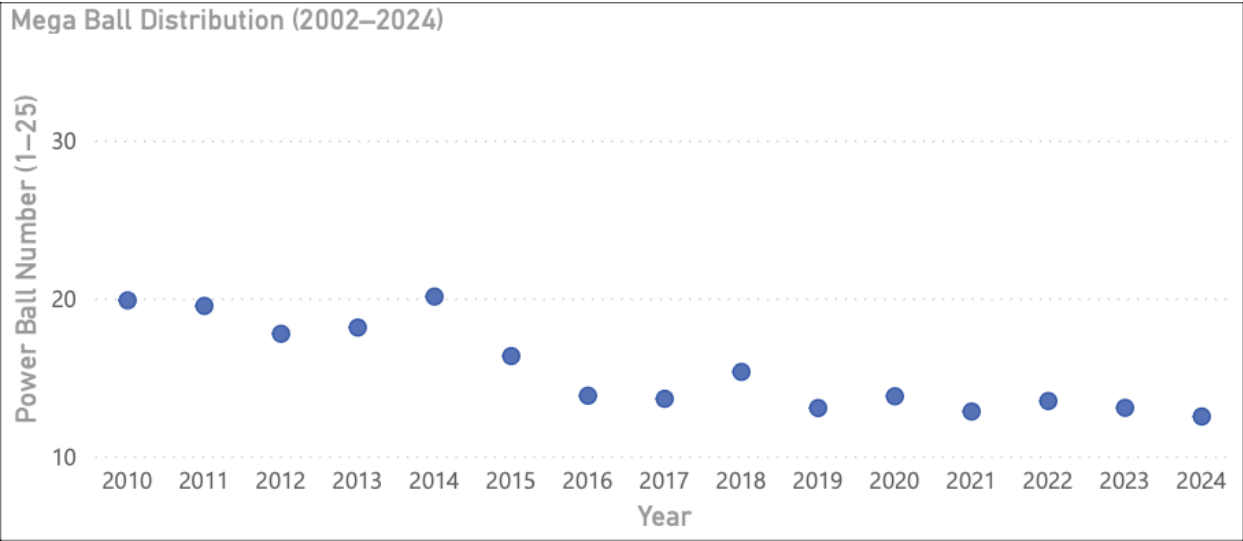


4. **Bubble Chart (Pick 10 Popularity):** Represents the **Law of Large Numbers**. With 13,000+ draws, the bubbles become uniform in size, demonstrating that given

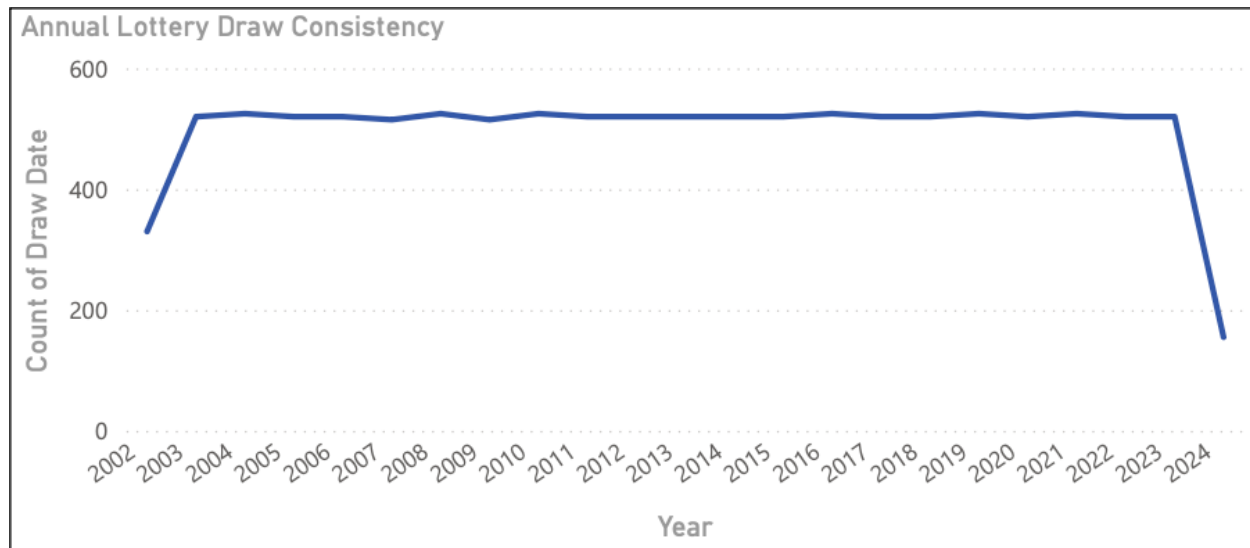
enough time, every number hits its exact mathematical probability.



5. **Density Plot (Special Ball Distribution):** Represents **Range Integrity**. It shows that the Mega Ball land across their entire 1–25 range without clumping, proving the machines are unbiased.



6. **Line Chart (The Heartbeat):** Represents **Systemic Consistency**. While the results are random, the steady volume of draws proves the lottery is a robotic, unending cycle.



Design Choices & Gestalt Principles

The dashboard utilizes the Principle of Proximity by grouping yearly histograms to highlight shifting patterns. Continuity is established through a consistent color palette, while Similarity in the bubble chart signals mathematical equality. To maintain clarity, the canvas height was extended to 1,500 pixels to create a vertical "scrolling story" that leads the viewer logically to the final recommendation.

Ethical Considerations

I used Power Query to unpivot the winning numbers; to mitigate the risk of misleading viewers about the volume of draws, I clearly labeled axes as "Frequency of Appearance." I assumed the 6th digit in Powerball represented the special ball, verified against official rules. The primary ethical risk—the "clustering illusion"—was addressed by including a flat trend line to visually confirm a lack of correlation over time. All data is public record from the NY Open Data portal, acquired ethically for the purpose of public transparency.